Devasheesh Mishra

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Summary

Founder and CTO of Recallr AI and AI/ML engineer specializing in conversational memory, retrieval systems, speech AI, and production LLM infrastructure; previously founded an acquired AI home-automation startup. Authoring three studies on scalable long-term memory, retrieval-free parametric memory, and hallucination dynamics in small language models. Selected for Founders, Inc.'s Canopy 2026 and Y Combinator's AI Startup School 2025.

Skills

Programming: Python, C/C++, JavaScript/TypeScript
AI/ML: PyTorch, Transformers, LLMs, Fine-tuning, LiteLLM, Pandas, NumPy, Matplotlib
Backend & Data: FastAPI, Pydantic, REST APIs, WebSockets, Gunicorn, Uvicorn, Alembic, PostgreSQL, MongoDB, Milvus, MinIO, Redis, Neo4j
Cloud & Infrastructure: AWS (ECS, ECR, EFS), Docker, Docker Compose, Kubernetes (K3s), Linux, Portainer, Traefik, Harbor, Terraform
Messaging: Kafka, RabbitMQ
Testing: Pytest (unit and integration testing), Selenium
Version Control: Git, GitHub
CI/CD & Observability: GitLab CI, GitHub Actions, Prometheus, Grafana, Loki, Promtail, Alertmanager, Elastic Stack (Elasticsearch, Kibana, Filebeat)
Embedded Systems: ESP-IDF, PlatformIO, Arduino, ESP32/ESP8266, ESP32-CAM, NodeMCU, Raspberry Pi

Work Experience

Founder and CTO — January 2025 – Present
Recallr AI Inc. — *San Francisco, CA, USA — On-site*

- Built Recallr, a persistent, queryable long-term memory layer for conversational AI systems that retains facts, preferences, relationships, and decisions across multiple conversations while preserving the source of each memory. Architected its ingestion pipeline, evolving knowledge graph, semantic retrieval, temporal reasoning, knowledge-update handling, and configurable merge and conflict-resolution rules.
- Evaluated Recallr on the LongMemEval benchmark, achieving 97.5% overall accuracy, including 97.0% temporal-reasoning accuracy and 97.4% knowledge-update accuracy. Delivered P95 latency of 408 ms for Low-Latency recall, 1.575 seconds for Balanced recall, and 8.619 seconds for Agentic recall.
- Evolved Recallr into the intelligence and memory layer for private capital, transforming fragmented deal memos, data rooms, diligence, meeting transcripts, partner notes, filings, returns, and investment-committee history into a continuously updated, queryable decision graph that preserves how a firm's investment judgment evolves over time.

Canopy 2026 — March 2026 – May 2026
Founders, Inc. — *San Francisco, CA, USA — On-site*

- Selected as one of 100 teams globally for Canopy 2026, a five-week on-site builder residency at Founders, Inc.'s San Francisco campus.
- Built and commercialized Recallr's memory layer for conversational AI systems during the program, acquiring multiple clients and generating approximately $15,000 in revenue.

AI/ML Engineer — September 2025 – November 2025
MIRA — *San Francisco, CA, USA — Remote*

- Real-Time Streaming Speech-to-Text Pipeline: Designed and implemented a low-latency, WebSocket-based streaming transcription system using Soniox STT models (stt-rt-preview-v2), with real-time speaker diarization, silence detection, and multilingual translation. Built a comprehensive benchmarking framework to evaluate ASR performance across Soniox, Deepgram, Google, and AssemblyAI using word error rate, latency, and speaker-attribution accuracy, then used the results to guide model selection and system design.
- Voice Fingerprinting and Speaker Verification: Engineered an end-to-end voice-biometric pipeline covering user enrollment, audio serialization, PostgreSQL storage, and real-time speaker identification. Used SpeechBrain's ECAPA-TDNN model trained on VoxCeleb to generate and compare speaker embeddings. Deployed the system as a standalone FastAPI microservice on AWS ECS, with parallel cosine-similarity verification across diarized speaker segments, identifying the enrolled user among ambient speakers with under 200 ms inference latency.
- Long-Term Memory Extraction from Personal Data: Built an LLM-powered memory-extraction pipeline using Gemini 2.0 Flash to process a user's Gmail corpus and identify durable signals such as personality traits, expertise, preferences, and behavioral patterns. Structured the extracted information into long-term user knowledge representations, enabling AI agents to maintain context across conversations.

Founder and CEO May 2024 – January 2025
Stapes AI *New Delhi, India — On-site*

- Founded an AI home-automation company, shipped v1.0 within one month, onboarded 50+ beta testers, and led the company through acquisition.
- Developed a home-automation platform integrating smart TVs, Fire TV devices, and switchboards, with control through voice and mobile apps; designed custom PCBs and ESP32 firmware for the device stack.
- Admitted to Buildspace's Nights & Weekends S5 program, backed by Y Combinator and a16z; subsequently white-labeled the home-automation technology for other operators.

AI/ML Engineer Intern August 2024 – September 2024
Proeffico Solutions Private Ltd. *Noida, India — On-site*

- Engineered and deployed a real-time shoplifting-detection system across retail locations, validating it with live on-site video feeds for a global client.
- Developed RDBMS Chat, an internal conversational AI tool that enabled non-technical staff to query complex relational databases in natural language, improving access to operational data and insights.
- Supported internal servers and virtual machines, helping maintain operational stability for backend systems across multiple projects.

Builder, Nights & Weekends S5 June 2024 – August 2024
Buildspace *San Francisco, CA, USA — Remote*

- Joined a startup program backed by Y Combinator and a16z, focused on rapid product development for Stapes AI.
- Accelerated Stapes AI through community feedback, mentorship, rapid prototyping, and successive product iterations.
- Applied Python, Flutter, and embedded-systems development with ESP-IDF to refine the product's software and hardware stack.

Technical Lead April 2024 – February 2026
GeeksforGeeks *New Delhi, India — On-site*

- Conducted hands-on machine-learning and deep-learning workshops for 80+ students, teaching core principles and practical applications.
- Organized and led 10+ GeeksforGeeks workshops across SRM, including coding-support sessions that reached more than 200 students.
- Managed eight technical contributors and project timelines while supporting troubleshooting and server operations for chapter initiatives.

Core Technical Team Member October 2022 – April 2024
GeeksforGeeks *New Delhi, India — On-site*

- Contributed as a core member while organizing the Phoenix Hackathon, where I also won first place.
- Organized Hack-Innovate, a two-day hackathon that attracted 300+ participants and showcased more than 50 project submissions.

Research

A Name Is Not a Memory: Relational Binding in Continual Parametric Memory for Conversational Agents *(In Progress)* August 2026 – Present

- **Goal:** Develop **retrieval-free** long-term memory for conversational agents by encoding user-specific knowledge in persistent LoRA adapters on a frozen language model; evaluate how parametric memory compares with retrieval-based systems such as Recallr AI on retention, identity binding, continual updates, and cross-user interference; and explore a scalable multi-LoRA architecture inspired by Mixture-of-Experts routing.

Hallucination Dynamics in Small Language Models: A Large-Scale Empirical Study *(In Progress)* January 2026 – Present

- **Goal:** Determine how sampling temperature affects hallucination behavior in small language models across model scales, quantization levels, and factual domains.
- **Approach:** Collected over 10 million data samples from SLMs including Qwen2.5-0.5B/1.5B/3B/7B, Qwen3-1.7B/4B/8B, Llama-3.2-1B/3B, Gemma-3-4B, and Mistral-7B. Tested $T \in [0, 1]$ at $\Delta T = 0.1$ on HaluEval-QA and True/False across politics, geography, history, science, health, cities, companies, facts, and other domains.
- **Result:** Identified a non-monotonic scale-temperature relationship: sub-1.5B SLMs often collapsed into one-class prediction priors at low temperature, so added sampling entropy could reduce hallucination error by weakening systematic bias; near the 1.5B competence boundary, higher temperature instead amplified factual instability, especially under 4-bit quantization, while many 3B–8B models remained comparatively stable. Effects varied by topic and were concentrated on marginal knowledge, showing that hallucination risk depends on the interaction of scale, calibration, quantization, and domain rather than temperature alone.

Recallr AI: Scalable Architecture for Dynamic Long-Term Memory in Conversational AI Agents *(In Progress)* April 2025 – Present

- **Goal:** Build scalable long-term memory for conversational AI that preserves evolving user state across sessions without adding latency to real-time responses.
- **Approach:** Built a versioned knowledge graph through decoupled asynchronous curation and synchronous retrieval. Its decision logic classifies new evidence as redundant, novel, additive, temporally superseding, expired, or directly conflicting; dual timestamps separate event time from discussion time, immutable version chains preserve state evolution, and a Git-inspired human-in-the-loop protocol resolves contradictions. Auto-Recall routes queries across Low-Latency, Balanced, and Agentic retrieval with adaptive graph traversal.
- **Result:** Benchmarked Recallr on the LongMemEval Oracle tier. Agentic Recall achieved 98.9% accuracy—32.7 percentage points above the nearest competitor—with 100% on knowledge updates, 99.2% on temporal reasoning, and 98.5% on multi-session questions; Low-Latency Recall retained 92.6% accuracy at 396 ms P95, approximately 4.5× faster at P95 than the fastest competing system.

Achievements

Y Combinator's AI Startup School: Chosen as one of 2,000 CS students worldwide to attend YC's first AI Startup School in San Francisco. (2025)

Google InnoSprint Hackathon: First runner-up. (2023)

Phoenix Hackathon: Won first place. (2022)

Code-A-Thon: First runner-up. (2022)

Live-Project Competition: Second runner-up. (2023)

Smart India Hackathon: First runner-up at the college level. (2023)

Certifications

Data Mining — *IIT, Kharagpur*

Neural Networks and Deep Learning — *DeepLearning.AI*

PyTorch for Deep Learning Bootcamp — *Udemy*

Extracurricular

Course Instructor — *September 2024 – April 2026*
GeeksforGeeks

Assistant Course Instructor — *July 2024 – September 2024*
GeeksforGeeks

Education

SRM Institute of Science and Technology — May 2022 – July 2026
Bachelor of Technology in Computer Science, Specialization in AI and ML; CGPA: 8.5 — *Chennai, India*